

ML systems

assumes the data
is given (exogenous)

control theory

assumes a known,
characterized plant

robotics / embedded

assumes a bounded
task and place

physical AI

where all three assumptions break at once

its own actions generate the data it learns from next

the model in the loop is learned, open-world, uncharacterized

it runs unbounded, in someone's home

a regime of its own, not the overlap of the three